SPE-230135-MS

Pioneering Advanced Ceramic Water Filtration Compact Skids for Efficient Water Injection in Tight Reservoirs: A Successful Pilot in Abduliyah Field

M. Al-Salal, S. Al-Mass, and K. Al-Abbasi, Kuwait Oil Company, Ahmadi, Kuwait; S. Kakade, N. Boareki, H. Al-Rashedi, B. Dashti, and N. Karadi, Baker Hughes, Ahmadi, Kuwait; D. Munck, T. Madsen, and M. Lydersen, LiqTech, Ballerup, Denmark

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Development of tight reservoirs under water flooding is considered as a challenging process in term of cost and effectiveness. A Potential problem associated with water flood application includes inefficient recovery due to poor injected water quality parameters in the selected reservoirs. Injecting water into tight reservoirs presents significant risks of **formation damage**, which can severely impair the effectiveness of the water injection process. To mitigate these risks, it's crucial to inject **high-quality water**.

The Risk of Formation Damage

Formation damage refers to the reduction in the reservoir's permeability around the wellbore, which restricts fluid flow. In tight reservoirs, this damage is particularly problematic because the already low permeability leaves little margin for error. The primary causes of this damage are:

- **Particulate Plugging:** Which is caused by presence of particulate solids having size greater than the pore throat of the target reservoir.
- **Clay Swelling:** Many tight reservoirs contain water-sensitive clays which are sensitive to contaminants present in the injected water
- **Scale Deposition:** Mixing injected water with incompatible formation water can lead to the precipitation of mineral scales, such as calcium carbonate (CaCO_3) or barium sulfate (BaSO_4). These hard, crystalline deposits can build up in the pore spaces, plugging them and reducing permeability.

Injecting water into tight reservoirs presents significant risks of **formation damage**, which can severely impair the effectiveness of the water injection process. To mitigate these risks, it's crucial to inject **high-quality water**.

The Need for High-Quality Water

To prevent the aforementioned formation damage, it is essential to inject high-quality water that is carefully treated and monitored. The key characteristics of high-quality injection water include:

- **Low Suspended Solids:** The water must be meticulously filtered to remove all suspended solids. This prevents particulate plugging and maintains good injectivity.
- **Chemical Compatibility:** The water's chemistry must be analyzed and adjusted to be compatible with both the reservoir's rock and the native formation fluids. This involves ionic content to prevent clay swelling and scale deposition.
- **Absence of Microorganisms:** Water must be filtered to remove bacteria that can grow within the well and reservoir. These microorganisms can produce slimes and other byproducts that plug pores, leading to what's known as "microbial-induced formation damage."

The success of water injection in tight reservoirs is highly dependent on controlling the quality of the injected fluid. Failing to do so can cause irreparable formation damage, leading to poor injectivity, reduced sweep efficiency, and ultimately, an economically unviable project.

Ceramic filtration technology is applied to demonstrate feasibility of water treatment & re-injection utilizing compacted skids and achieving treated water quality across ranges of feed water types/sources as part of various initiatives in the fields of IOR/EOR in compliance with KOC's 2040 strategy for sustained production and reserves growth.

The Abduliyah field of Kuwait, the characteristic of injection water requires a certain high specification, to avoid inefficient sweep and eliminate possible reservoir damage leading to injectivity loss, wellbore plugging, and by-pass oil.

The method demonstrates the feasibility of water treatment & re-injection compacted skid system to treat water with TDS ranges between 120,000 mg/L to 240,000 mg/L, temperature range from 10°C to 90°C and achieving permeate/treated water quality across ranges of feed water types/sources.

This paper presents the implementation of first successful use of advanced membrane filtration and media cleaning protocols in Abduliyah field of Kuwait, demonstrating performance restoration across multiple cleaning cycles, including backwash and chemical clean-in-place (CIP) procedures. The system design also incorporated corrosion control, scale inhibition, and biological fouling prevention through optimized chemical dosing, complemented by inert gas blanketing to eliminate dissolved oxygen ingress. Removal of bacteria and iron was consistently achieved.

The study further investigates the injection performance into target reservoirs using three different water sources—which includes aquifer, tertiary treated water, and tetra treated water by evaluating injection rates, pressure behavior, impact on formation in terms of skin build up, injectivity deterioration and compliance with both stringent and relaxed water quality criteria. A modular relaxation approach enabled systematic assessment across different injection phases, including relaxed and bypass filtration modes.

Injection into the Middle Marrat reservoir was executed using three distinct water sources: Zubair aquifer water from a production well located 9 km from the pilot site, and tertiary and tetra-treated Kabed water from remote preliminary treatment facilities. The pilot results validates the feasibility of deploying advanced ceramic membrane filtration technology at field level, effectively de-risking waterflooding operations in tight reservoir environments.

Pre-defined injection water quality was derived based on the core flood and compatibility studies. Advanced ceramic membrane ultrafiltration ensured TSS < 0.2 ppm, non-detectable oil-in-water, and 100% removal of particles ≥1 micron in size. Dissolved oxygen was maintained below 10 ppb in over 95% of samples. This marks the first surface and sub-surface oil and gas application of such high-spec water quality, establishing a milestone in advanced ceramic filtration for tight reservoir development.

Introduction

Development of tight reservoirs under water flooding is considered as a challenging process in term of cost and effectiveness. Oil was discovered in the Abduliyah (AB) field in 1984. The MMR in the Abduliyah field is a significant Jurassic carbonate reservoir with an average porosity of 8-12%, a permeability ranging from 1-10 MD, and a total thickness of 400ft. These Jurassic reservoirs are crucial for Kuwait due to their additional light oil reserves.

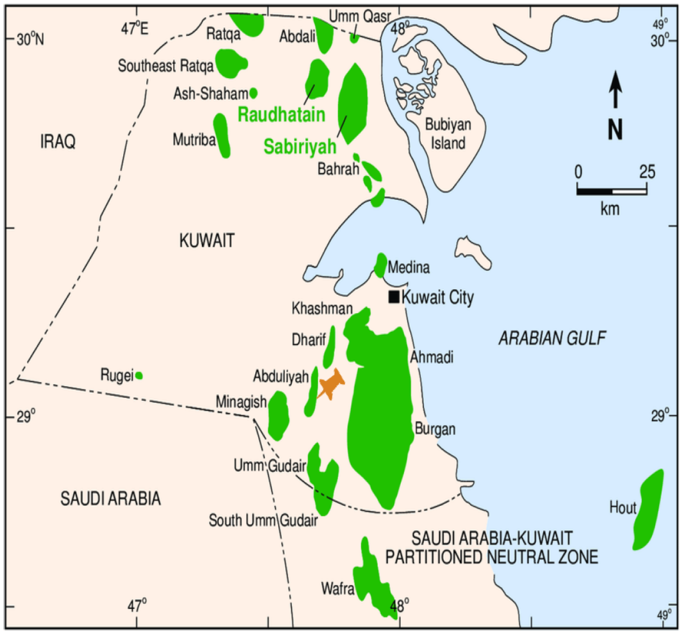

Figure 1—Pilot plant location

The oil accumulation in these reservoirs is a result of both primary and secondary porosity, with secondary porosity sometimes linked to a fracturing system. However, the heterogeneity of the carbonate reservoirs, along with high pressure and the difficulty of intersecting fractures, have made drilling and production challenging and expensive. Due to prolonged production from Middle Marrat reservoirs, the formations are getting depleted and pressure maintenance through water injection/water flood is very essential to ensure the continuity of production and oil recovery from these reservoirs.

A potential problem associated with water flood application include inefficient recovery due to injected water quality parameters in the selected reservoirs. In West Kuwait, the characteristic of injected water requires a certain high specification, in order to avoid inefficient sweep and eliminate possible reservoir damage which could eventually leads to injectivity lose, wellbore plugging, and by-pass oil.

A foundational core study by the Kuwait Oil Company (KOC) on its carbonate reservoirs has led to a critical recommendation of injecting high-purity water as essential for the long-term health and reservoir management.

Current industry practices for water injection in tight reservoirs have several gaps, primarily related to filtration effectiveness, operational efficiency, and the long-term impact on the reservoir. These gaps highlight areas where new technologies, like ceramic membranes, are needed.

- In effective Filtration and Formation Damage: A major gap is the inadequate removal of fine particles from injection water using conventional methods. Conventional filters often have inconsistent pore sizes, allowing small, suspended solids and oil droplets to pass through. When this water is injected into tight reservoirs, these particles can clog the tiny pore throats, leading to formation damage. This damage reduces the injectivity of the well, requiring higher injection pressures and ultimately leading to a lower sweep efficiency and less oil recovery.

- Fouling and Maintenance Challenges: Conventional filters, are highly susceptible to fouling. Oily components, waxes, and other organic matter in produced water can quickly coat the membrane surfaces, drastically reducing the filtration rate (flux). This requires frequent and intensive chemical cleaning or filter replacement, leading to increased operational downtime and maintenance costs. The lack of robust, long-lasting filtration media is a significant operational and economic gap.
- Limited Performance in Harsh Conditions: Produced water is often hot, acidic, or contains corrosive substances. Most polymer membranes and traditional filtration media cannot withstand these conditions, leading to rapid degradation and failure. This limits their applicability in many oilfield environments, forcing operators to use more complex, expensive, and energy-intensive pre-treatment processes.
- High Energy Consumption and Environmental Impact: Many conventional water treatment methods are energy-intensive and may use chemical additives. This not only increases the operational cost but also raises environmental concerns. The current industry practice often falls short in providing a cost-effective and environmentally sustainable solution for treating and reinjecting produced water, especially when dealing with large volumes. This leads to a larger carbon footprint and regulatory challenges.

A turnkey project was designed to provide services of skid mounted water treatment system at West Kuwait assets in order to treat three types of water; aquifer water (i.e. produced water), effluent water and tertiary treated waste water sources before re-injecting at target locations. Well-01 of Abduliyah field, is the 1st well amongst several pilot tests planned by KOC, targeting Middle Marrat and other formation in West Kuwait.

Superiority of Ceramic Membrane Filtration over Conventional Methods

Ceramic membrane filtration is superior to conventional methods for water injection in tight reservoirs because it provides a more effective and sustainable solution. Unlike conventional methods that often struggle with consistent pore size and mechanical stability, ceramic membranes offer highly uniform pores, which prevents fine particles from clogging the narrow pore throats of the reservoir rock. This precise filtration is crucial for maintaining injectivity and preventing formation damage. Furthermore, ceramic membranes are chemically and thermally stable, withstanding harsh conditions often present in produced water, such as high temperatures, acidity, and oily components. This robustness results in a longer operational lifespan and reduced maintenance costs compared to polymer membranes and other conventional filters, which degrade more quickly. Consequently, the reliability and efficiency of ceramic membrane technology lead to improved sweep efficiency and enhanced oil recovery from challenging, low-permeability reservoirs.

The Critical Role of High-Purity Water Injection in Improved Oil Recovery

The injection of high-purity water into hydrocarbon-bearing formations is a fundamental practice in Improved Oil Recovery (IOR) operations, particularly in waterflooding. This practice is not merely a preference but a critical requirement for maximizing hydrocarbon recovery, maintaining reservoir integrity, and ensuring operational efficiency. The significance of high-purity water can be attributed to several key factors:

- Prevention of Formation Damage and Enhanced Injectivity- Impurities present in injection water pose a substantial threat to reservoir permeability and injectivity. These impurities include:

a. Suspended Solids: Particulates such as rust, silt, sand, and biological matter can physically plug the pore throats of the reservoir rock. This formation damage significantly impedes the flow of injected water and, consequently, the displacement of oil, leading to reduced sweep efficiency.
b. Scale-Forming Ions: Waters containing ions like calcium, barium, strontium, and sulfate can precipitate as mineral scales (e.g., calcium carbonate, barium sulfate) within the reservoir pores and production tubing. Such scaling reduces effective permeability, restricts fluid flow, and can damage production equipment.
c. Corrosion Products: Corrosive elements in the injected water can lead to the formation of iron oxides and other corrosion products that, similar to suspended solids, can plug the formation and reduce injectivity.
d. Emulsifying Agents and Oil-Wetting Compounds: Residual hydrocarbons or other organic contaminants in the injected water can promote the formation of undesirable emulsions within the reservoir or alter the rock's wettability from water-wet to oil-wet. A water-wet state is generally more conducive to efficient oil displacement. Utilizing high-purity water effectively mitigates these risks, preserving the reservoir's natural permeability and ensuring consistent, unimpeded water injection.

- Optimization of Sweep Efficiency and Displacement- Maintaining the injectivity and integrity of the reservoir allows for more uniform and controlled water movement through the formation. This leads to an optimized sweep efficiency, where the injected water contacts and displaces a greater volume of crude oil from the reservoir matrix. In advanced IOR/EOR techniques, such as polymer/chemical flooding, the purity of the injection water is even more paramount. Impurities can degrade the chemical additives (e.g., polymers), reducing their designed viscosity-enhancing properties and compromising mobility control, which is crucial for efficient oil displacement.
- Mitigation of Microbial Contamination and Reservoir Souring- Injected water can serve as a conduit for microorganisms and nutrients into the reservoir environment. The proliferation of these microbes can lead to:

a. Biofilm Formation: Microbial biofilms can accumulate and plug pore spaces, further exacerbating injectivity issues.
b. Hydrogen Sulfide (H_2S) Production: Sulfate-reducing bacteria (SRB) are a significant concern, as they can generate highly corrosive and toxic hydrogen sulfide gas. This phenomenon, known as reservoir souring, poses substantial safety risks, increases infrastructure corrosion, and necessitates processing steps for the produced hydrocarbons.
c. High-purity water, often coupled with appropriate biocide treatments and de-aeration, significantly reduces the introduction and growth of these problematic microbial populations.

- Preservation of Equipment Integrity- The presence of impurities in injection water can cause accelerated erosion, corrosion, and scaling in surface facilities (e.g., pumps, pipelines, filtration units) and downhole equipment. This results in increased operational downtime, elevated maintenance costs, and reduced equipment lifespan. The use of high-purity water substantially lessens the wear and tear on critical infrastructure, contributing to more reliable and cost-effective operations.

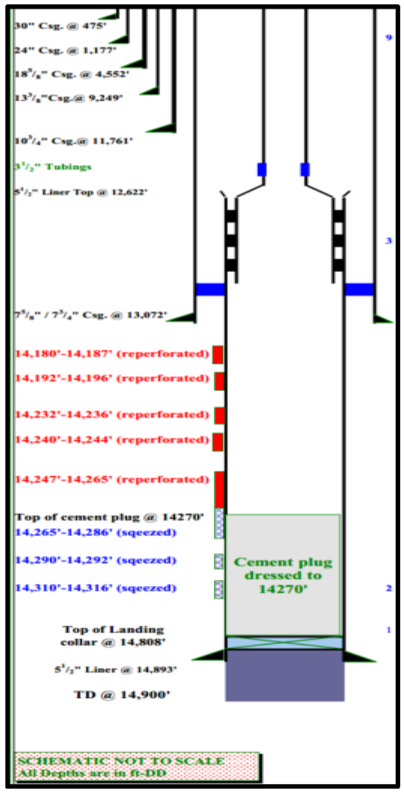

Figure 2—Targe Well-01 Schematics

Processes to Achieve High-Purity Water for Injection

To ensure the high-purity water required for injection, a multi-stage treatment process was employed:

- Prefiltration: Mesh filters were used to filter out the coarser particles.
- Ceramic Filtration: This crucial step was carried out to achieve a remarkable level of purity by the use of tubular ceramic membrane employing a crossflow filtration technology.
- Deaeration: This process was undertaken to remove dissolved gases by the use of pre feed tank. To further ensure the absence of oxygen, oxygen scavengers were utilized. Additionally, nitrogen blanketing was implemented to preserve the water quality throughout the treatment and storage phases by preventing atmospheric oxygen from re-entering the system

The comprehensive approach to water treatment and its recommendation underscore a proactive strategy to ensure the sustained productivity and longevity of the carbonate reservoirs minimizing potential damage and securing long-term operational success.

Description of the Pilot Facility

To address the stringent water quality requirements, advanced filtration technologies has been deployed. The surface facility comprises of a state-of-the-art filtration system designed to meet both routine operational demands and more flexible conditions, utilizing ceramic membrane technology.

This system leverages the capabilities of both hybrid Silicon Carbide (SiC) and Zirconia Ultrafiltration (UF) membranes and pure Silicon Carbide (SiC) Microfiltration (MF) membranes to achieve remarkable water quality.

The key objective of the Pilot Project -

- To evaluate and understand water flooding application response in tight reservoirs in term of effectiveness, recovery factor and cost.
- Explore the performance of targeted reservoirs against the aquifer, effluent and tertiary treated water injection rate, pressure and water specifications relaxed in terms of TSS and particles removal.
- Familiarize operator with aquifer, effluent and tertiary water treatment & re- injection compacted skids system technology.
- Developing a workflow that enables optimization of water injection schemes may lead to a more complete approach to maximizing asset value and have a better decision making for future reservoir development.
- Demonstrate feasibility and do-ability of aquifer water, effluent water & tertiary treated water further treatment and re-injection utilizing compacted skids system achieving permeate water quality for the given range of feed water qualities.
- Demonstrate effective membrane filtration media cleaning procedure; including performance restoration after multiple cleaning cycles (i.e., backwash, chemical ‘clean in place (CIP)’, etc.) and protect the injection system from corrosion, scale, bacterial growth with suitable chemical dosing and avoiding ingress of dissolved Oxygen with inert gas blanketing as required.
- Demonstrate system robustness against upset conditions e.g. high TSS carry- over, slug of excessive solids, overdosing of production chemicals, inadvertent /incorrect "clean in place’, backwash operations, etc.
- Establish optimum operating parameters (notably flux, recycle rate, recovery percentage, operation robustness, sustainable well head pressure, etc.) for aquifer water, effluent water & tertiary treated water re-injection compacted skids systems, specific to the aquifer composition, filtration type (membrane / other filtration media), filter feed pump, chemical dosing etc.
- Demonstrate feasibility of aquifer water treatment & re-injection compacted skids system to treat aquifer water TDS ranges between 120,000 mg / L to 240,000 mg:/ L and temperature range from 10 °C ta 90 "C and achieving permeate water quality across ranges of feed water qualities

Pilot Plant Design

Injection into the Middle Marrat reservoir was executed using three distinct water sources:

- Zubair Aquifer water from a production well located 9 km from the pilot site
- Tertiary-treated water from remote treatment facilities and
- Tetra-treated Kabed water from remote treatment facilities.

This filtration plant boasts a modular and containerized design, allowing for easy mobilization and demobilization between different well locations/sites. This inherent flexibility makes it an ideal solution for various operational needs. The fit for purpose facility design was selected for this pilot as outlined in below Table 1 and Table 2

Water Treatment Facility Design Criteria

Table 1—Feed Water Parameters

Feed		
Flowrate	3,000 BWPD (Max.) 1,000 BWPD(Min.)	7-day average based on sum of feed pumps flow data
TSS	<200 mg/L	2 Samples pr. day
Oil In Water (OIW)	100 mg/L	2 Samples pr. day
Dissolved Oxygen	<5000 ppb	2 Samples pr. day
pH range	5 – 9	2 Samples pr. day
Temperature	10 – 50 °C	2 Samples pr. day
Variation in inlet water		
TSS	< 1000 mg/L	Max. 1 peak pr. week

Table 2—Permeate water parameters

Permeate		
Flowrate	> 99 % of feed flowrate	7-day average based on feed- and permeate flow data
TSS	< 0.2 mg/L	2 Samples pr. day
Oil In Water (OIW)	0.0 mg/L	2 Samples pr. day
Particle Size Removal	≥ Particles size removal≥ 95% of particles of size ≥ 0.07 µm < 0.10 µm Implies More than 95 % of the particles from the feed ranging between 0.07 micron to 0.10 micron are removed. ≥ 98% of particles of size ≥ 0.10 µm implies more than 98% of particles from the feed with size greater than 0.10 micron are removed. 100% of particles of size > 1.00 µm implies All particles greater than 1 micron are removed.	Relative to particle content in Feed.
Total Iron	All particle-bound Fe or FeS will be removed under the above particle size conditions.	

A Comprehensive Introduction to Filtration Membrane

Membrane filtration is broadly divided into distinct categories. This often involves a multi-stage approach. Among these, microfiltration (MF), ultrafiltration (UF), and nanofiltration (NF) are distinct membrane processes, differentiated primarily by the size of the pores in their membranes and, consequently, the types of contaminants they can remove.

- Microfiltration (MF) uses membranes with relatively larger pores (typically 0.1 to 10 micrometers) to physically separate suspended solids, bacteria, and some larger colloids from the water.
- Ultrafiltration (UF) employs membranes with finer pores (typically 0.01 to 0.1 micrometers) than microfiltration. This allows UF to effectively remove not only suspended solids and bacteria but also viruses, proteins, and larger organic molecules.
- Nanofiltration (NF) features even smaller pores (typically 0.001 to 0.01 micrometers), positioning it between ultrafiltration and reverse osmosis in terms of filtration capability. NF membranes can reject multivalent ions (like hardness-causing calcium and magnesium) and larger organic molecules, while allowing smaller monovalent ions (like sodium and chloride) to pass through.

Table 3—Filtration Types

Filtration type	Pore size (Micrometer)	Description
Microfiltration	0.1-10	Effective at removing suspended solids, large bacteria, and some viruses, but smaller dissolved substances and most viruses can pass through.
Ultra filtration	0.001- 0.1	UF removes everything that MF does, plus it can filter out colloids, proteins, and most viruses
Nano filtration	0.001- 0.01	It filters out everything that UF and MF remove, along with most organic matter and multivalent ions (ions with a charge greater than +1 or –1)

Media availability in the market

The material of construction (MOC) for a filter is selected based on the specific operational requirements of the filtration process. Common materials include alumina, zirconia, titania, and silicon carbide. Additionally, composite materials can be engineered to combine the advantageous properties of different constituents, thereby enhancing overall media performance. The characteristic of each material is listed in Table 4.

Table 4—Membrane MOC and their advantages

Membrane type	Characteristics
Alumina	• Good Mechanical Strength • Chemical resistance across wide pH range
Zirconia	• Excellent chemical resistance • High thermal stability
Titania	• Good chemical stability • Photocatalytic property
Silicon Carbide	• Good Mechanical Strength • High Thermal Resistance • High Chemical Resistance • Good Hydrophilicity

Membrane stacking utilized in pilot plant

The pilot plant utilizes cylindrical membrane filters, each measuring 1.2 meters in length. Each housing unit accommodates 137 membrane rods, resulting in a total of 411 membrane rods per skid (3 housing units per skid). The pilot plant incorporates three skids in total. Two of these skids are configured to achieve standard water purity parameters, while the third skid is set up for replacement with other module to deliver water meeting relaxed purity specifications. Each individual membrane rod provides a filtration surface area of 0.33 m^2, contributing to a total operational filtration surface area of approximately 406 m^2 for the entire pilot treatment facility.

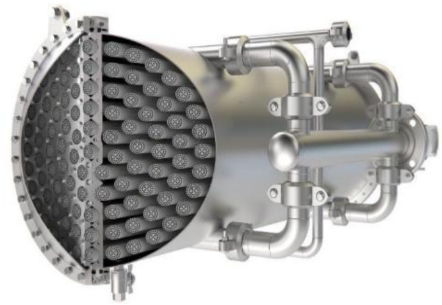

Figure 3—Membrane Housing

Operating Parameters of the filtration unit

Based on the core flood analysis recommendations two filtration parameters were employed under this project, parameters for each mode are detailed below:

Normal Operating Parameters

For routine operations requiring the highest purity, the system utilizes advanced hybrid membranes: The parameters targeted while filtration was being carried out at normal parameters are listed in the Table 5.

Table 5—Details of Membrane used to achieve normal operation parameters

Membrane type	**Characteristics**
Membrane Type	Hybrid SiC and Zirconia Ultrafiltration (UF)
Pore Size	60 nanometers
Oil in Water (OIW)	0.0 PPM
Total Suspended Solids (TSS)	< 0.2 PPM
Particle removal	100 % Removal for particles above 1 micron

To achieve these stringent normal parameters, fit for purpose hybrid technology membranes (HTM) of Silicon Carbide (SiC) and Zirconia Ultrafiltration (UF) were manufactured (refer Figure 4).

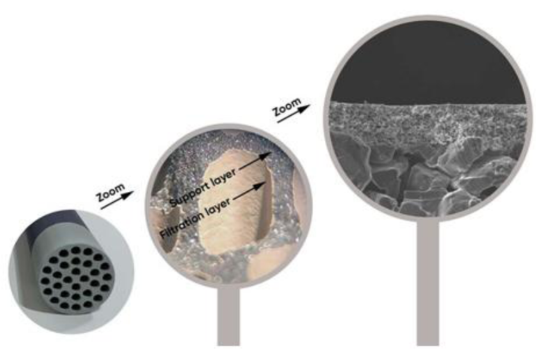

Figure 4—HTM membrane

These membranes have an incredibly small pore size of 60 nanometers. Ultrafiltration is a pressure-driven membrane separation process that removes suspended solids, bacteria, and other pathogens, making it suitable for high-purity water requirements. The combination of SiC and Zirconia offers enhanced durability, chemical resistance, and filtration efficiency.

Relaxed Operating Parameters

The relaxed parameters were specifically derived from core flood analysis to evaluate the reservoir's response to injection of varying water quality and to optimize treatment costs without compromising reservoir health. To achieve the objective of the relaxed water injection, the system utilizes robust microfiltration (MF) membranes.

The parameters targeted while filtration was being carried out at relaxed parameters are as listed in Table 6.

Table 6—Details of Membrane used to achieve Relaxed Operating Parameters

Parameter	Value
Membrane Type	Pure Silicon Carbide (SiC) Microfiltration (MF)
Pore Size	200 nanometers
Oil in Water (OIW)	1 PPM
Total Suspended Solids (TSS)	< 1 PPM
Particle Removal	100% removal for particles above 2 microns

For these relaxed parameters, the project employed Pure Silicon Carbide (SiC) Microfiltration (MF) membranes. These membranes have a larger pore size of 200 nanometers. Microfiltration is also a physical filtration process that removes suspended solids, bacteria, and larger colloids.

This dual-membrane approach provides unmatched adaptability and performance to meet the pilot objective, advancing water purity standards in demanding industrial environments.

Pilot plant process description

Pre-Filtration and Degassing

The incoming water to the facility is directed to the Pre-Feed Tank, also known as the surge tank. This tank serves a critical role in degassing the water, preparing it for subsequent filtration steps. The pre-feed tank features a two-chamber design, where water flows from the first chamber to the second via an overflow mechanism, facilitating thorough degassing.

From the pre-feed tank, a dedicated pump transfers the water to Pre-Filters. These filters are engineered to remove coarse particles from the feed water, utilizing a 200-micron filtration size. This is to ensure no damage is caused by membrane by coarse particles entering the main filtration. The filtered water is then stored in the Feed Tank, which is blanketed with nitrogen to prevent oxygen ingress and maintain water quality. Refer Figure 5

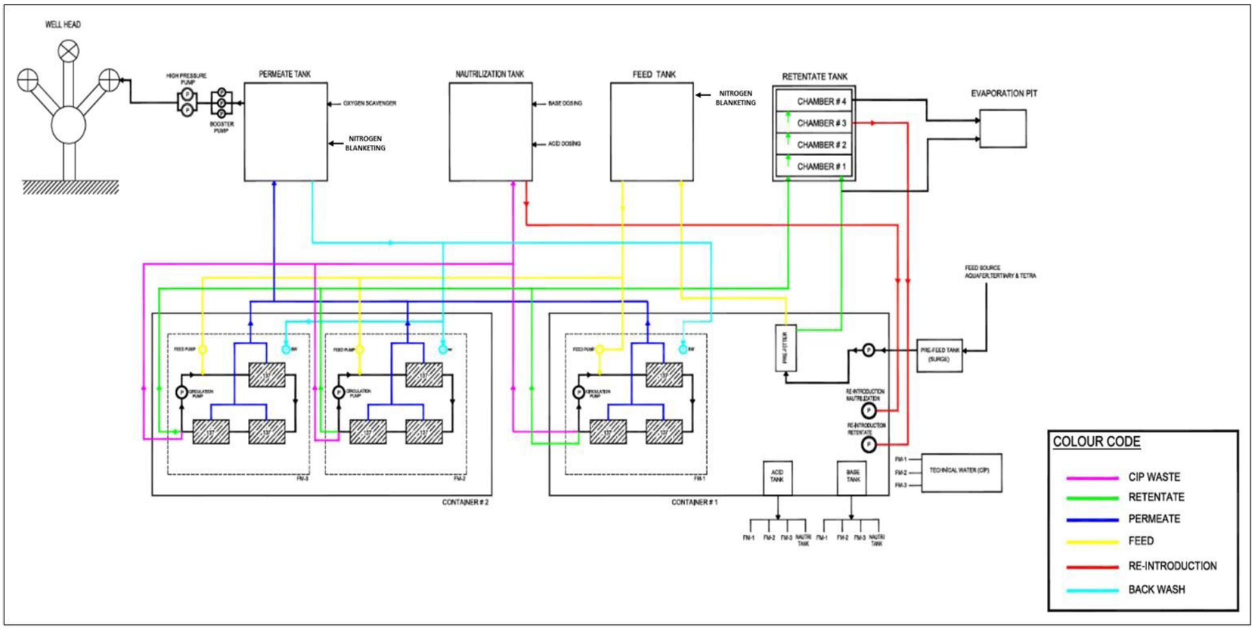

Figure 5—Plant process flowchart

Main Ceramic Filtration and Health Monitoring

The water from the feed tank is subsequently directed to the main Ceramic Filters. The water treatment module consists of three membrane housings in the same crossflow loop, each with 137 tubular membrane rods for a total of 411 tubular membranes. Continuously the feed is led to the crossflow loop while the retentate is bled from the crossflow loop.

The housings are connected in series as to obtain desired crossflow capacity. A single crossflow pump delivers the continuous cleaning effect, a so-called washing, through shear forces on the membrane surface caused by the large crossflow velocity.

A feed pump delivers the operating pressure. This pressure difference across the membranes, Trans Membrane Pressure (TMP), drives cleaned water (permeate) through the membranes.

A backwash pump is periodically engaged to reverse the TMP in one housing at a time. These forces clean permeate back through the membrane pores for a cleaning effect.

Concentrated CIP chemical can be dosed directly into the crossflow loop and spend CIP solution leaves the crossflow loop for neutralization.

Crossflow Ultrafiltration general process description

Feed enters the crossflow-loop (refer Figure 6) with a relatively small flowrate. The crossflow circulates feed with a very large flowrate through the internal tubular membrane channels. The large crossflow velocity ensures a substantial flushing of the membrane surface that significantly reduces fouling.

The feed pressure in the crossflow-loop drives water through the membrane surface and thus clean water (permeate) emerges on the outside of the cylindrical membrane. Several membrane rods are combined in housings where permeate is collected. A bleed with a minor flowrate is led from the crossflow-loop as up-concentrated, dirty, water (Retentate).

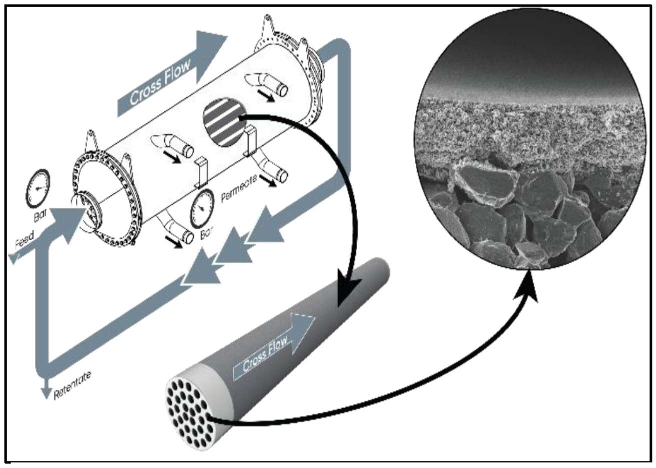

Figure 6—Crossflow

Fouling management system employed for Ceramic membrane filters

Fouling management within the filtration system is systematically addressed through a multi- pronged approach, primarily leveraging three distinct methods. The principal method for fouling management is crossflow. During the continuous filtration process, the feed water is strategically flowing tangentially across the surface of the filtration media. This dynamic flow generates a shear force that effectively dislodges particles attempting to block the membrane pores, thereby maintaining system efficiency. This process is continuous and integral to the ongoing filtration operation. Throughout this process, the Transmembrane Pressure (TMP) is meticulously monitored as a critical operational parameter.

In addition to continuous crossflow, periodic backwashing serves as a secondary, yet crucial, fouling mitigation technique. During a backwash cycle, the normal filtration process is temporarily halted. The flow direction of the permeate water is then reversed, causing it to flow back through the ceramic filter media and out through the membrane pores. This reverse flow effectively dislodges accumulated particles and restores the permeability and overall health of the ceramic membranes.

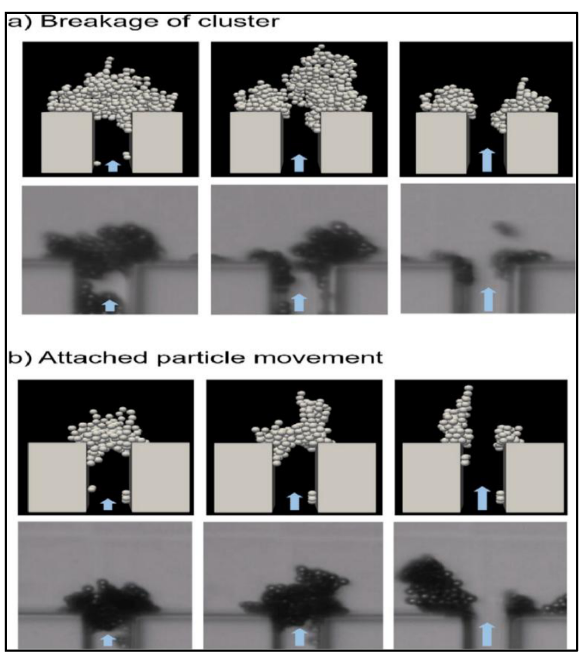

Figure 7—Fouling Management

The **Transmembrane Pressure (TMP)** is a key operational indicator, representing the differential pressure between the external and internal pressures across the ceramic filter media. It provides a real-time assessment of the degree of fouling present within the filter media. When the TMP surpasses a predetermined set point, a **Chemical Clean-in-Place (CIP)** procedure is automatically initiated.

The CIP sequence commences with the evacuation of the existing process water from the filtration system, followed by the introduction of technical water (potable water) for an initial flushing of the filters. Upon completion of this flushing phase, a base solution is introduced into the filters. This base solution is then circulated for a predefined duration, which is determined by the quality of the feed water and the severity of fouling. This circulation effectively cleans the filters. Subsequently, an acid solution is introduced into the circulation loop, undergoing a similar circulation process as the base. Following both the base and acid circulation phases, technical water is again introduced to thoroughly flush the system, thereby concluding the CIP operation and effectively restoring the optimal performance of the ceramic filter media.

Table 7—Summary of CIP frequency during pilot operation

CIP Frequency During Pilot Operation (17th Jul 2022 to 25th Jan 2023)		
	No. of Acid CIPs	No. of Base CIPs
Filter module 1	7	7
Filter module 2	7	6
Filter module 3	4	4

Neutralization Operation

The spent acid and base solutions utilized for cleaning the ceramic filters are discharged into a dedicated neutralization tank. The pH within this tank is continuously monitored in real-time. Based on the prevailing pH, either acid or base is precisely dosed into the tank to maintain the pH within a near-neutral range, specifically between 6 and 8.

A sufficient resting period to ensure a stable and consistent pH, once a neutral pH is confirmed, the neutralized solution from the tank is reintroduced into the filtration system, where it is combined with the incoming feed water. This integrated recycling process for the used chemicals significantly contributes to a near-zero waste output from the filtration facility.

Retentate Management and Water Recycling

The reject stream from the membrane filters containing a high concentration of the filtered-out particles and oil is directed towards the retentate tank. This tank (Figure 9) is uniquely designed with four chambers. Water flows sequentially from the first chamber to subsequent chambers via an overflow mechanism. The initial two chambers are specifically engineered for the settling of filtered particles at the bottom of the tank. After the settling process is complete in these chambers, the water flows to the third chamber. In the third chamber, any oil floating on the top layer of the water is directed to the fourth chamber. Oil from the fourth chamber is then directed to a nearby waste pit and disposed of following all environmental protection guidelines.

Settled sediments in the first and second chambers are periodically drained and directed to the waste pit using dedicated pumps.

Crucially, water from the third chamber of the retentate tank is reintroduced into the filtration process. This is achieved by passing it through the pre-filters and subsequently storing the filtered water in the feed tank. This recycling and reintroduction mechanism significantly minimizes discarded water, ensuring a remarkable recovery ratio with 99% utilization of the feed water flowing to the filtration plant.

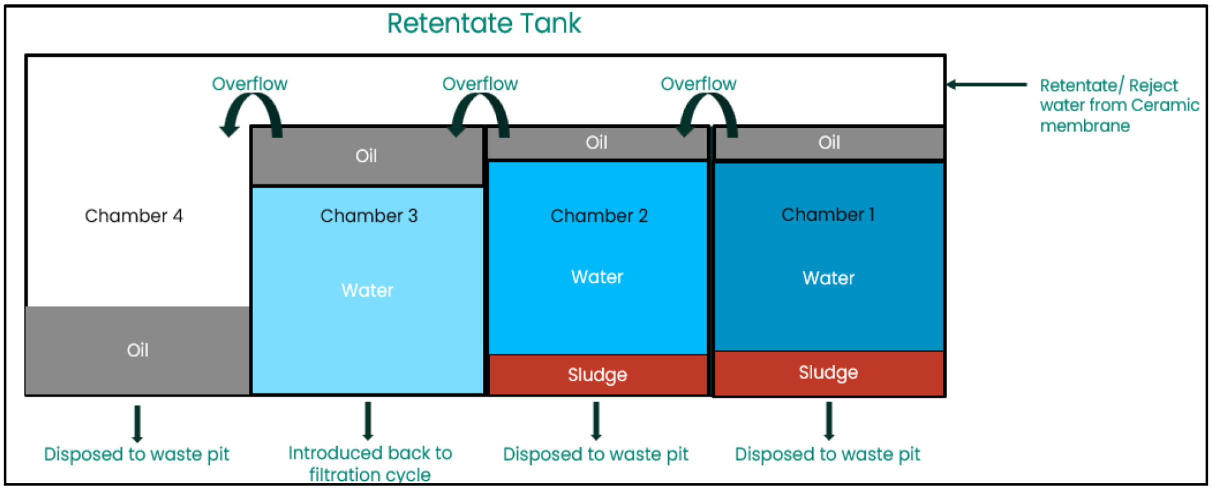

Figure 9—Retentate tank General Arrangement

Advantages of using ceramic membrane technology

Ceramic membranes offer superior flux and permeate recovery due to their unique properties, leading to higher capacities than conventional methods. This is attributed to:

Factors Contributing to High Capacity

Elevated Porosity: A high density of interconnected pores maximizes volumetric flow, leading to greater permeation flux per unit area.

- Optimized Pore Sizes in Support/Carrier Layer: The support layer provides mechanical integrity and minimizes hydraulic resistance, preventing bottlenecking and maximizing throughput.
- Extremely Low Contact Angle to Water (Exceptional Hydrophilicity): A contact angle of 0.5 degrees for SiC signifies extreme hydrophilicity, offering (Figure 10):
  - Profoundly Reduced Fouling Propensity: A stable hydration layer acts as a barrier, preventing foulant adhesion and growth
  - Maximized Water Permeation: Water molecules readily enter pores with minimal energetic barrier, maximizing flux rates.
  - Minimized Resistance to Water Transport: The combination of porosity, optimized support pores, and low contact angle significantly reduces hydraulic resistance, leading to higher filtration capacity for a given pressure

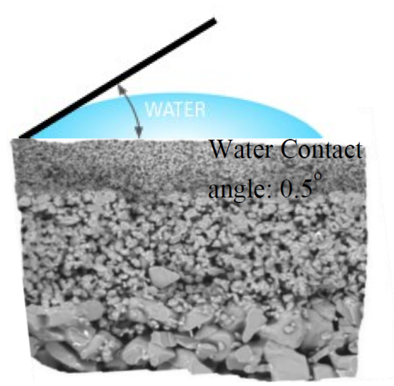

Figure 10—SiC contact angle to Water.

Operational and Economic Benefits

- Maximized Filtration Capacity per Unit Area
- Reduced System Footprint
- Decreased Energy Consumption
- Reduced Backwash Water Consumption
- Lowered Capital and Operational Expenditures for Ancillary Components

In essence, the high capacity of membranes stems from their engineered porous architecture, exceptional hydrophilicity (0.5-degree contact angle), and the resulting minimal hydraulic resistance.

Pilot Operation Execution

The injection plan was prepared and revised by KOC based on the preliminary injectivity test after in the well. Original expected injection rate was envisaged as max. 1500 b/d.

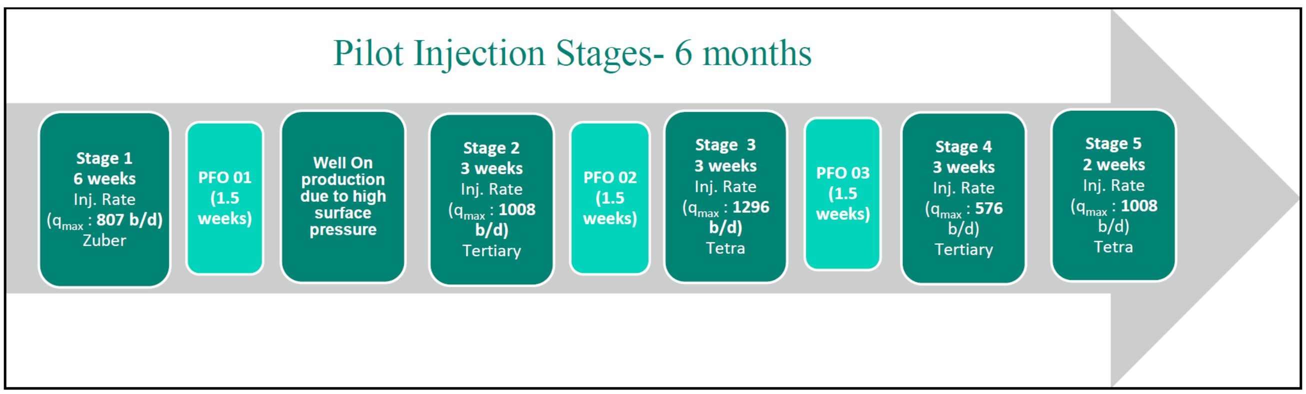

Figure 11—Injection Plan

Table 8—Summary of pilot parameters for each injection stage

Stage	Source Water	Max Inj. Rate, b/d	Max Pressure, psi	Filter Size, Micron	TSS, ppm	OiW, ppm	DO, ppm	Target Parameter
1	Zuber Original	807	3350	One	0.18	0	0	TSS< 0.2 ppm OiW = 0 PSD removal > 1 micron DO
2	Tertiary Original	1008	4668	One	0.02	0	0	
3	Tetra Original	1296	4765	One	0	0	0	
4	Tetra Relaxed	576	3300	Two	1	0.2	0	TSS< 1 ppm OiW < 1 ppm PSD removal > 2 micron DO
5	Tertiary Relaxed	1008	4171	Two	0.4	0.03	0	

Injection Summary of the Target Well-01

An injection operation was conducted in the target well-01 utilizing three distinct water sources across five sequential stages (refer to Figure 12). The plot represents the reservoir behavior after treatment and re-injection in each stage of injection including aquifer water, tertiary treated water and tetra treated water. A total of 74,197 barrels of treated water was injected during the entire operation. The overall injectivity was found to be good as there were no signs of any plugging or injectivity loss within the reservoir from the surface pressure trends throughout the whole duration of pilot, which confirms the effectiveness of filtration. Several PFO tests were performed for skin evaluation in between stages ensuring no adverse damage on the targeted reservoir.

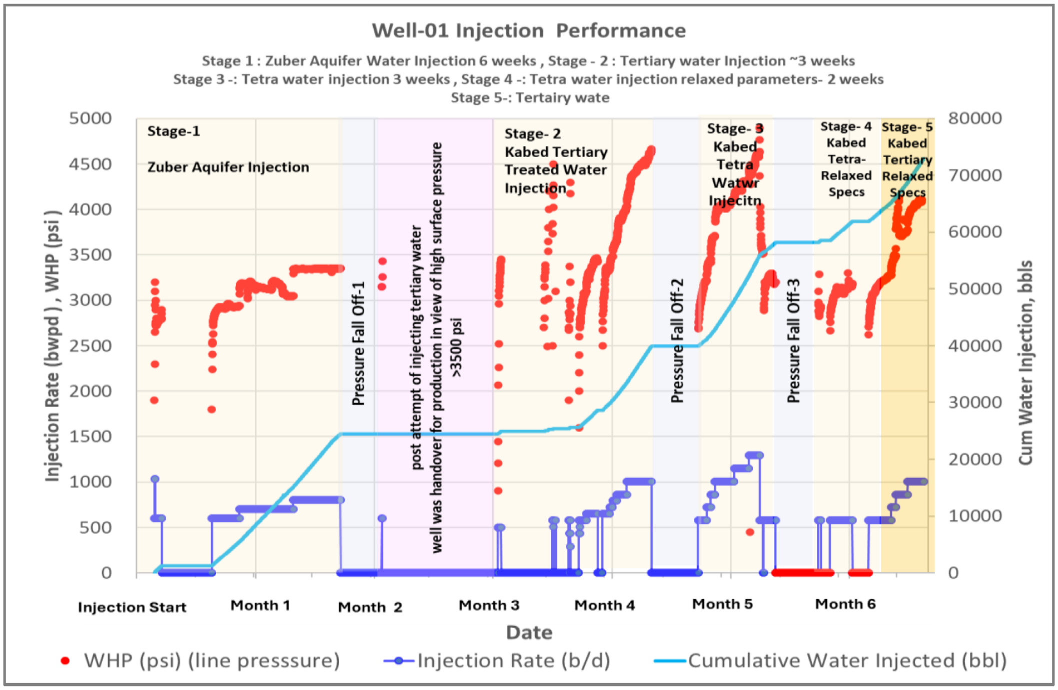

Figure 12—Injection parameters over the course of 5 stages of injection

Stage 1: This initial stage involved the injection of Zuber aquifer water after treatment under normal operating parameters. The injection rate varied from 600 to 800 BPD over a six-week period. A stable wellhead pressure of 3345 psi was consistently observed. Following the completion of Stage 1, a 10-day Pressure Fall-off (PFO) test was performed for skin evaluation.

After stage 1, tertiary treated water injection was attempted, however the surface pressure reached to envisaged design capacity – 3500 psi. The tertiary water salinity was very low ~ 400-500 ppm and the resultant column pressure in the well was lower than the high salinity Zuber aquifer water with salinity of 240,000 ppm. In other words to inject tertiary water high pressure pump with surface pressure limit up to 5200 psi was required (~5200 pressure at surface equivalent to max. allowable frac pressure for the formation). It is important to note that in both cases the bottom hole pressure is constant and is irrespective of the salinity of injected fluid.

Stage 2: This stage utilized tertiary treated water injected under normal parameters. The injection rate in this stage reached a maximum of 1050 BPD, with the wellhead pressure not exceeding 4688 psi. A 10-day PFO test was conducted upon completion of this stage for skin evaluation purposes.

Stage 3: During this stage, tetra treated water was injected under normal parameters. The maximum injection rate achieved was 1296 BPD, with a peak wellhead pressure of 4226 psi. A 10-day PFO test followed this injection stage.

Stages 4 and 5 were dedicated to analyzing reservoir compatibility through the injection of water under relaxed parameters. Two water sources were selected for these stages: tetra treated water and tertiary treated water.

Stage 4: This stage involved injecting tetra treated water under relaxed parameters. A maximum injection rate of 576 BPD was achieved, and the wellhead pressure reached 3300 psi during the injection period.

Stage 5: In the final stage, tertiary treated water was injected. A maximum injection rate of 1008 BPD was attained, and the peak wellhead pressure observed was 4171 psi.

Results

The overall water quality in terms of outlet TSS, OiW and particle size removal shows more than 98 % of the sample size achieving the desired water quality and meeting the success criteria of the pilot outcomes. A bi-weekly 3rd party sample testing was carried out to ensure the validation of the water quality achieved during the entire pilot operation as shown in Figure 13 and Figure 14

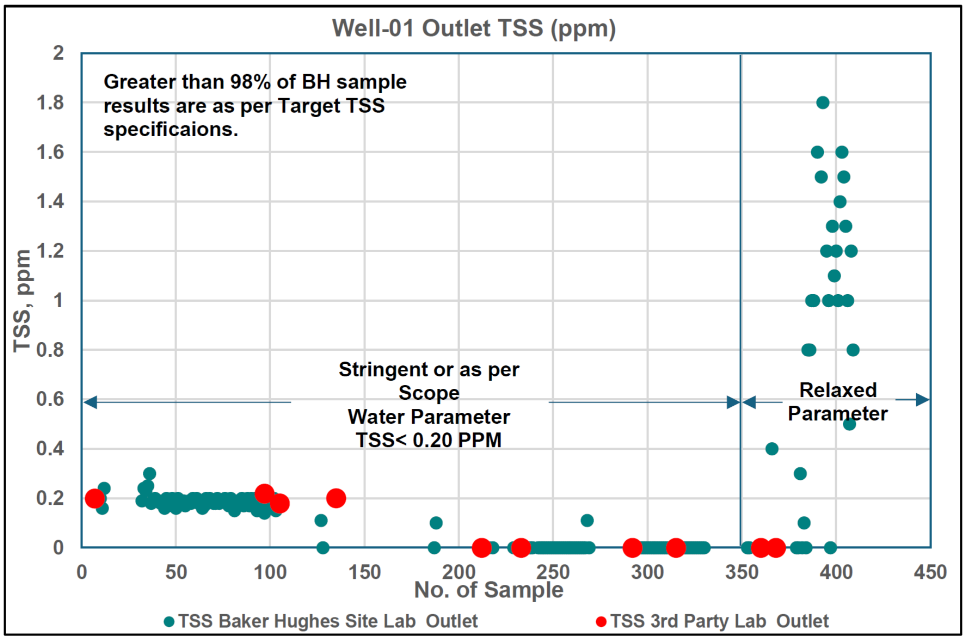

Figure 13—Outlet Water TSS, pilot site laboratory and 3rd party laboratory

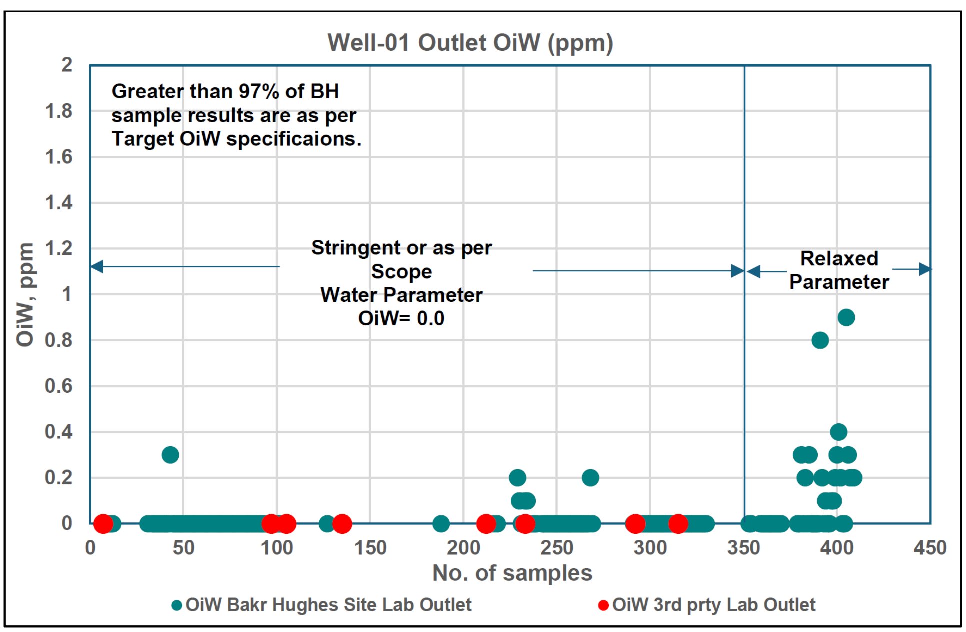

Figure 14—Outlet Water, pilot site laboratory and 3rd party laboratory

A Particle size distribution (PSD) analysis had been performed to establish particle size distribution before and after treatment of all water types throughout the duration of pilot using a Malvern PSD analyzer. From the Figure 15, all particles from the feed greater than 1 micron was removed in the first stage (i.e. aquifer in normal conditions), meeting target specification of normal conditions which consists of 100% removal of particles >1, Meanwhile, in the later stages (tertiary and tetra in both normal and relaxed conditions), all particles from the feed had been removed completely as per project scope (Relaxed conditions ~ 100% removal of particles >2).

Outlet water quality

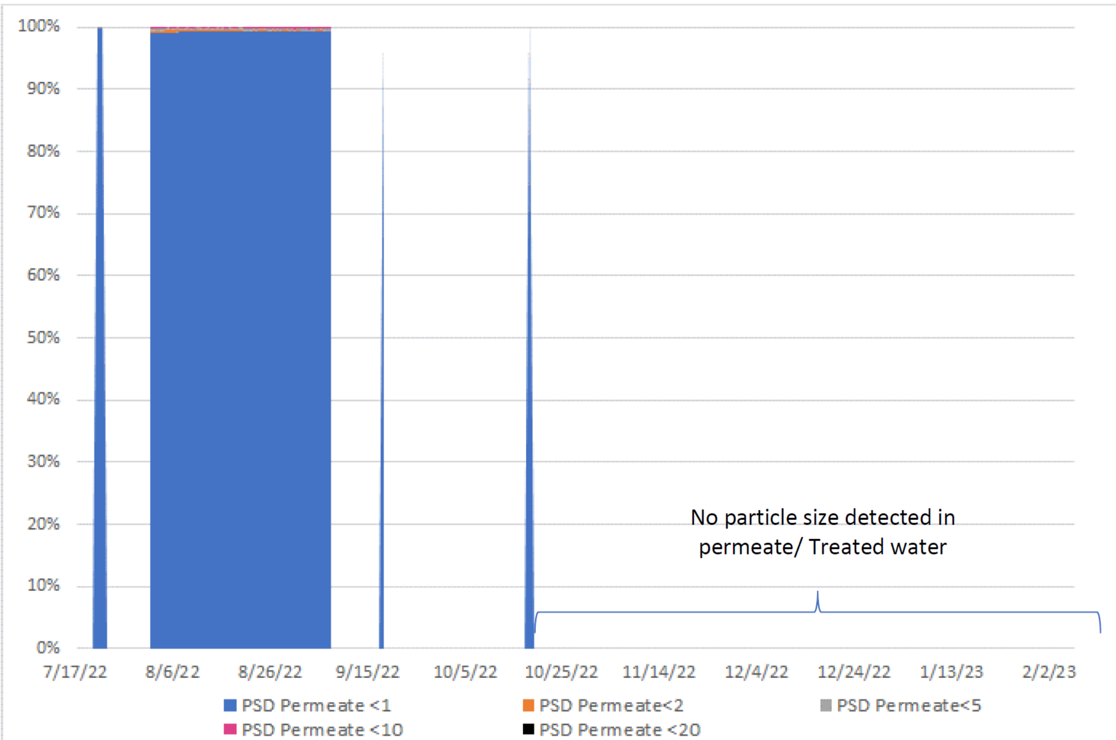

Figure 15—Outlet Particle Size- Pilot Site Laboratory

Scale up potential of Ceramic filtration technology to full field. Having successfully completed the pilot field operation, the scale-up potential of ceramic membrane filtration to full-field application for water injection in tight reservoirs is highly promising. The demonstrable performance in a controlled pilot setting, where the technology has proven its ability to maintain injectivity and handle the specific challenges of produced water in a real-world environment, provides a strong foundation for expansion. This successful pilot indicates that the technology's benefits, such as superior filtration efficacy, robustness against harsh conditions, and reduced fouling tendencies, can be replicated across a larger operational footprint. Scaling up would involve deploying multiple ceramic filtration units or larger modules, integrating them into existing or new water treatment infrastructure, and optimizing their operation to meet the increased demand for injection water. Furthermore, the long-term stability and lower lifecycle costs observed in pilot studies suggest that a full-field implementation could lead to significant improvements in overall field economics and sustainability, driving enhanced oil recovery across the entire reservoir

Conclusion

- Pilot Execution was with LTI free operation, zero waste site, an excellent efficiency > 99 % was achieved No downtime from Baker Hughes demonstrating operational excellence in project management.
- Abdullyaha Middle Marrat Reservoir Pilot the First Pilot of Skid Mounted Water Treatment System at WK Asset project under Master Technology Agreement was completed within stipulated time frame achieving all the success criteria for the HSE, target water quality, recovery ratio, operational efficiency, injection rates and pressures, injection of 3 different types of source water.
- The pilot results shows encouraging outcomes for achieving the water injectivity in Middle Marrat reservoir. The maximum rate achieved during the pilot operation is 1296 b/d. The maximum surface pressure observed is 5000 psi.
- The operational efficiency/facility uptime was 99 % throughout the pilot operation.
- High recovery ratio of 99% was achieved, keeping very minimal discharge – waste water
- The pressure fall off test conducted after each stage of source water type and filtered water quality shows no plugging and injectivity loss.
- The offset producer wells shows considerable production gain due to increased pressure by efficient injection.
- The application of ceramic membrane technology for improved oil recovery in tight reservoirs proves to be beneficial in large scale field level applications due to its robustness, long life, minimal chemical requirements, higher recovery ratio, less downtime for back wash (Back wash in seconds) and CIP (Cleaning in Place), compatibility with high and low salinity water, high temperature of the source water.
- The surface facility injection pumps for the large sale applications to be designed considering the fact that different salinity water will results in different surface pressure for a constant injection pressure at sand face.
- The pilot results validates the feasibility of deploying advanced ceramic membrane filtration technology at field level, effectively de-risking waterflooding operations in tight reservoir environments.

Abbreviations

IOR : Improved Oil Recovery
EOR : Enhanced Oil Recovery
TDS : Total dissolved solids
CIP : Clean In Place
TSS : Total Suspended Solids
OiW : Oil in Water
PPM : Part per Million
PPB : Part per Billion
MMR : Middle Marrat
MD : Measured Depth
SRB : Sulfate reducing bacteria
UF : Ultra filtration
MF : Micro Filtration
NF : Nano Filtration
HTM : Hybrid Technology Membrane
SiC : Silicon carbide

BWPD : barrel of water per day
MOC : Material of construction
TMP : Trans membrane pressure
CAPEX : capital expenditure
OPEX : operating expenditure
PFO : Pressure Fall Off

References

1. Rasha Al Moraikhi, Nitin Kulkarni, Dipak P. Patil, Mahesh Sounderrajan, Naveen K. Verma, Riyad Quttainah, Eman Al-Kandari, and Mohannad Al-Mohailan Kuwait's First Successful Implementation of MPD in Deep Jurassic Reservoirs of Abdulyiah Field in West Kuwait. Paper SPE-188649-MS presented International Petroleum Exhibition & Conference held in Abu Dhabi, UAE, 13-16 November 2017. https://onepetro.org/SPEADIP/proceedings-abstract/17ADIP/17ADIP/D021S032R001/200357
2. Final well report for Well -XX (Abduliyah): *Provision & Services for Skid Mounted and Mobile Water Treatment System at WK Asset* by Baker Hughes, LiqTech in February 2023.
3. Final KOC Project Report: *Provision & Services for Skid Mounted and Mobile Water Treatment System at WK Asset*, by Baker Hughes, LiqTech December 2024.
4. Fuyuhiko Ishikawa, Satoru Mima, Arata Nakamura. – Industrial Scale Demonstration of Ceramic Membrane Filtration for Produced Water Treatment. SPE paper SPE-196160-MS presented at the SPE Annual Technical Conference and Exhibition, Calgary, Alberta, Canada, September 2019. https://doi.org/10.2118/196160-MS
5. I. Dunsmuir, G. Guthrie, T. Cottrell, A. Nijmeijer, B. Gydesen Reck – The Deployment of Ceramic Membrane Technology to Treat Stable Oil in Emulsions in Produced Water Offshore. SPE paper SPE-186131-MS. presented at the SPE Offshore Europe Conference & Exhibition, Aberdeen, United Kingdom, September 2017. https://doi.org/10.2118/186131-MS
6. Pierre Pedenaud, Evans Wayne, Heng Samuel, Bigeonneau Didier – Ceramic Membrane And Core Pilot Results For Produced Water Management. SPE paper OTC-22371-MS presented at the OTC Brasil, Rio de Janeiro, Brazil, October 2011. https://doi.org/10.4043/22371-MS
7. F. Al Salem, T. Thiemann, K. Kawamura, A. Nakamura, V. Poulose, H. Saibi – Purification of Produced Water from Kuwaiti Oil Fields Using Ceramic Membranes Paper Number: IPTC-24571-MS presented at the International Petroleum Technology Conference, Dhahran, Saudi Arabia, February 2024. https://doi.org/10.2523/IPTC-24571-MS